

Yun Qiao

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EDUCATION

The Ohio State University

Bachelor of Science in Mathematics, Minor in Computer Science

GPA: 3.962/4.0 | Honors Program Student

Columbus, OH

Expected May 2027

RESEARCH INTERESTS

- **Large-Scale Computational Optimization:** Second-order and quasi-Newton methods, randomized sketching, and efficient algorithms for PDE-constrained optimization and high-dimensional machine learning.
- **Numerical Linear Algebra & Iterative Solvers:** Preconditioning, Krylov subspace methods, and efficient sparse linear algebra for PDE solvers and large-scale optimization.

RESEARCH EXPERIENCE

Quasi-Newton Methods for Large-Scale PDE-Constrained Optimization

Research Assistant | Department of Mathematics, The Ohio State University

Jan 2026 - Present

Advisor: Prof. Anna Yesypenko

- Investigate quasi-Newton methods for large-scale PDE-constrained optimization within a variational framework.
- Study randomized sketching approaches to reduce computational cost while preserving numerical accuracy in PDE-constrained optimization.

LLMAdBench: A Human Preference Benchmark for Advertising in LLM Responses

Research Contributor | The University of Texas at Dallas

May 2026 - Present

Advisor: Prof. Chonghuan Wang, Naveen Jindal School of Management

- Contributed to a benchmark examining how advertisement placement and sponsorship disclosure affect human preferences for LLM-generated responses.
- Conducted pairwise preference annotation across six dimensions covering user experience, advertiser value, and overall preference.
- Helped construct a benchmark containing 18,576 dimension-level human judgments from 258 source conversations across four advertisement placements and two disclosure conditions.

Topological Data Analysis (TDA) for Equity Markets

The Ohio State University

Sep 2025 - May 2026

Research Mentor: Luči Krnić, Ph.D. Candidate

- Investigated relationships among financial assets using a retrospective Python pipeline combining Ward hierarchical clustering, return normality tests, and PCA visualization.
- Computed rolling-window Vietoris-Rips persistent homology using correlation-based distances; extracted separate H_0 and H_1 persistence landscape L_1 amplitudes to summarize changes in asset topology over time.
- Generated topology time series, persistence diagrams, and statistical visualizations for retrospective analysis of financial market structure.

Code: github.com/Yun-Qiao11283/TDA

RESEARCH MANUSCRIPT

LLMAdBench: A Human Preference Benchmark for Advertising in LLM Responses.

Co-author. Manuscript under review at ICLR 2027.

HONORS AND AWARDS

- **Dean's List:** Five consecutive semesters (Autumn 2023, Spring 2024, Autumn 2024, Spring 2025, Autumn 2025, Spring 2026).
- **Mathematics Department Scholarship**
- **Academic Scholarship (First Prize)**

SELECTED COURSEWORK

Operations Research and Statistics

- Linear Optimization (ISE 5200) - Graduate level
- Mathematical Statistics (STAT 4201/4202)
- Topics in Risk Modeling II (MATH 5638; machine learning) - Graduate level
- Numerical Linear Algebra (MATH 5603) - Graduate level

Advanced Mathematics

- Honors Analysis I and II (MATH 4181H/4182H) - Graduate level
- Honors Linear Algebra and Differential Equations (MATH 5520H) - Graduate level
- Honors Complex Analysis (MATH 5522H) - Graduate level
- General Topology and Knot Theory (MATH 5801) - Graduate level

Computer Science

- Data Structures in C++ (CSE 2122)
- Discrete Structures (CSE 2321)
- Data Structures and Algorithms (CSE 2331)
- Programming in C++ (CSE 1222)

SKILLS

- **Programming and Tools:** Python, C++, GitHub
- **Optimization Solvers:** Gurobi, CPLEX
- **Numerical Methods:** Quasi-Newton methods, randomized sketching
- **Data Analysis:** Persistent homology, persistence landscapes, hierarchical clustering, PCA, rolling-window analysis
- **LLM Benchmarking:** Pairwise preference annotation, benchmark dataset construction